Hsuan-Fu (Jeff) Hsu

AI and ML Engineering | Retrieval, RAG and Model Evaluation

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EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science | GPA: 3.6/4.0 — Jan 2026 - May 2027 (Expected)

National Taipei University — New Taipei City, Taiwan
Bachelor of Science in Computer Science | GPA: 3.5/4.0 — Sept 2020 - June 2024

SKILLS

Languages: Python, TypeScript, C++, C#, SQL

ML and Evaluation: PyTorch, Sentence Transformers, TensorFlow, NumPy, tsfresh, LOSO validation

LLM and RAG: FAISS, Ollama, OpenAI Agents SDK, LangChain, LangGraph, FastAPI

Cloud and Testing: Docker, Google Cloud Run, pytest, Vitest, xUnit, GitHub Actions, Git

EXPERIENCE

National Taipei University — New Taipei City, Taiwan
Undergraduate Researcher | Advisor: Prof. Bor-Shing Lin — July 2022 - Feb 2024

- Built a **Python** pipeline for noninvasive uric acid estimation from 30 participants and two wavelengths, applying Bessel filtering, quality checks, and tsfresh feature extraction.
- Compared five ML model families and four DL architectures using **LOSO cross validation**; the team's linear regression model achieved 10.3% MAPE with 10 selected features.
- Received individual NSTC undergraduate research funding and earned fourth place in NTPU's project competition with a team of three.

Tech-Management Workshop Co., Ltd. — Taichung, Taiwan
Software Engineer Intern — July 2026 - Aug 2026

- Independently built a government application with **ASP.NET Core MVC** and **SQL Server**, deployed across 29 offices with status tracking and incremental data synchronization.
- Reduced database latency by at least 50% and overall resource usage by at least 30% through data processing and storage improvements.

PROJECTS

Scientific Retrieval Fine-tuning — Personal Fork — Oct 2026
Python, PyTorch, Sentence Transformers | GitHub

- Fine-tuned **MiniLM** with **PyTorch** and Sentence Transformers, improving mean nDCG@10 by 3.2% and Recall@10 by 2.1% over random-negative training on 300 SciFact test claims across three seeds.
- Built a controlled evaluation workflow with data leakage checks, false-negative review, 53 unit tests, and model save/reload verification.
- Extended the retriever into a **FastAPI** RAG service (FAISS, local LLM via Ollama) and tested 6 setups on 300 test claims: a reranker raised nDCG@10 from 0.645 to 0.695, but blinded human checks found 79–84% faithful answers in every setup.

Promptfoo Agent Retry Validation — Personal Fork — Oct 2026
TypeScript, Vitest, OpenAI Agents SDK | GitHub

- Added nested retry policy error paths using **TypeScript**, with six regression cases failing before the change and passing after it.
- Added 81 test cases with **Vitest** and the OpenAI Agents SDK, including 60 configuration checks and 11 retry behavior scenarios with mocked model clients.

Classroom AC Setpoint Simulator — USC IEEE HackIoT (team of 4) — Oct 2026
TypeScript, Next.js, React, Three.js | GitHub

- Ported a teammate's Python AC setpoint model to **TypeScript** and matched its output hour by hour over 1,344 hours of data.
- Built a **Next.js** simulator that shows each hour's setpoint, energy use, and bill; in a simulated 8-week run, one study room used an estimated 27% less energy (2,143 kWh) than a fixed 22°C setting.
- Ran **YOLO** people counting on a Raspberry Pi 4 with teammates at about 7 frames per second and sent the counts to Supabase.

PUBLICATION

Coauthor, "Noninvasive Portable System With Dual Near-Infrared Light Technology for Monitoring Uric Acid Concentration," IEEE Transactions on Human-Machine Systems, accepted for publication, Aug 2026.